

Prep Test 2: Functions and Polynomial Graphs

75 minutes | 100 points | 10 questions

Name: _____

Date: _____

Directions

- Show your work.
- Use interval notation where requested.
- Questions 6 and 8 require qualitative polynomial sketches. Exact extrema and a uniform vertical scale are not required.

Use the space provided; additional work may go on the back.

Question 1

10 points

1 Problem 1

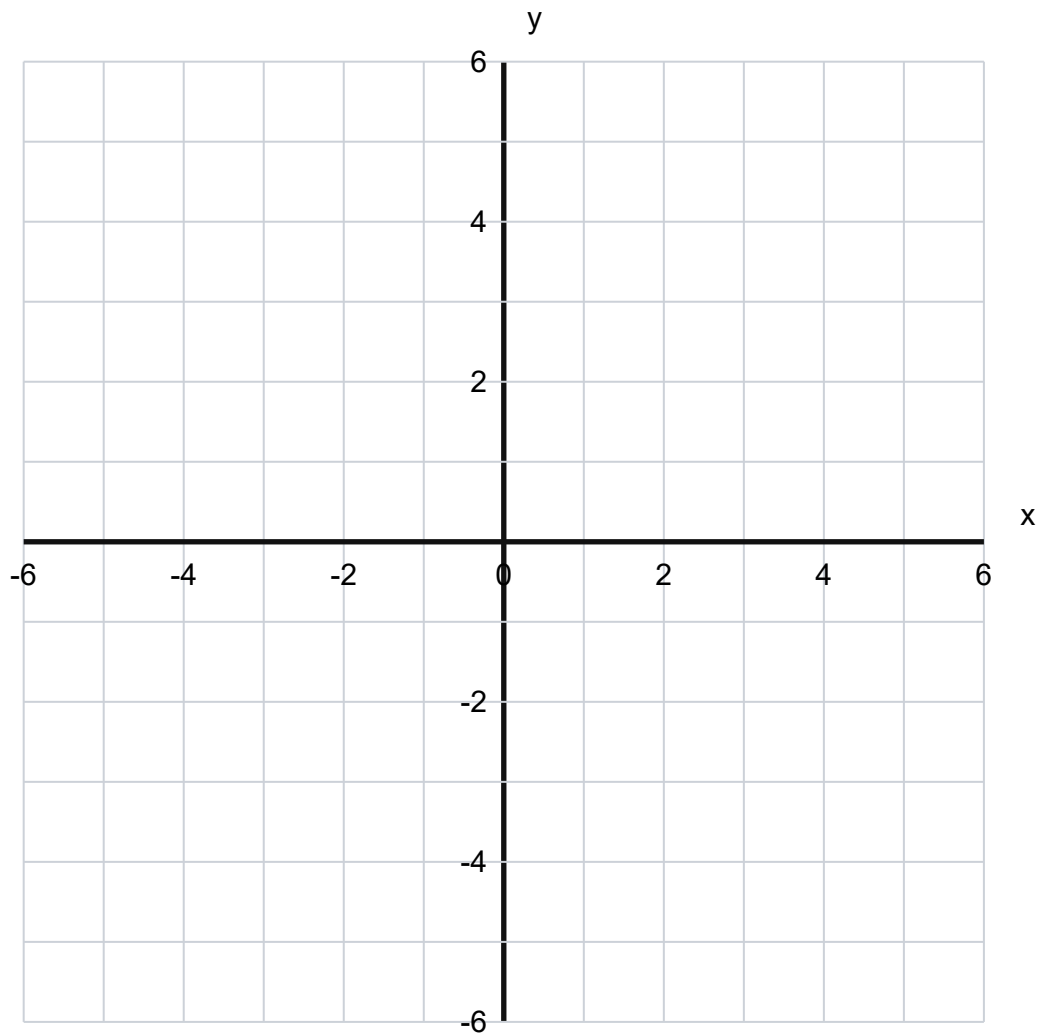
10 points

Let $p(x)=2x+4$ for $x<-1$, and $p(x)=3$ for $x\geq-1$.

- (a) Find $p(-2)$, $p(-1)$, and $p(2)$.
- (b) Graph p on the provided coordinate plane.
- (c) Give the domain and range in interval or set notation.

Question 1: your graph

One unit per square.



Answers and work

Question 2

12 points

2 Problem 2

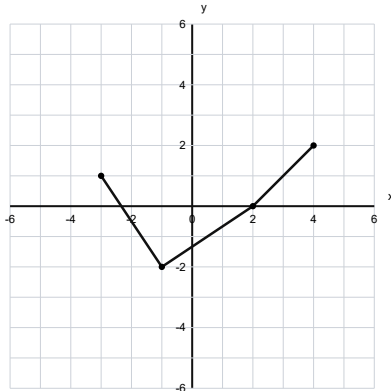
12 points

The graph of f is shown below. Its vertices are $(-3,1)$, $(-1,-2)$, $(2,0)$, and $(4,2)$, joined by straight segments, with included endpoints. Let $h(x) = -f(x - 2) + 1$.

- (a) Graph h on the provided coordinate plane. Label its vertices.
- (b) Give the domain and range of h .

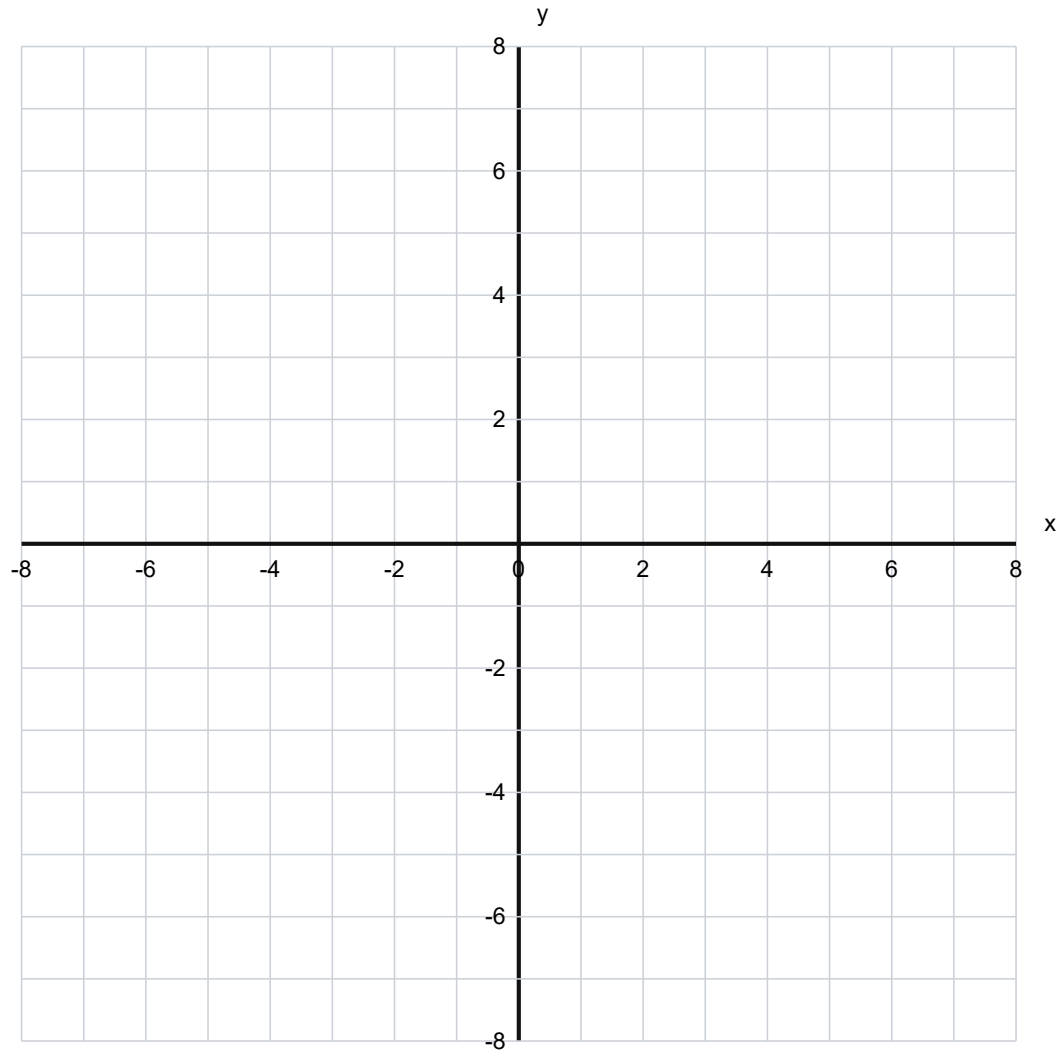
Given graph: f

Work space



Question 2: your graph

One unit per square.



Answers and work

Question 3

14 points

3 Problem 3

14 points

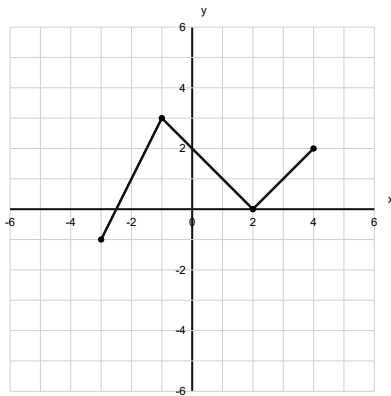
The graph of f consists of straight segments through $(-3,-1)$, $(-1,3)$, $(2,0)$, $(4,2)$, in that order, with included endpoints. Let $g(x) = 1 - \frac{3}{5}x$.

(a) Graph $f(g(x))$ and $g(f(x))$ on the provided coordinate plane. Use a solid line for $f(g(x))$ and a dashed line for $g(f(x))$. Label the vertices and both graphs.

(b) Give both domains and describe the transformations of f in each composition. Show your work.

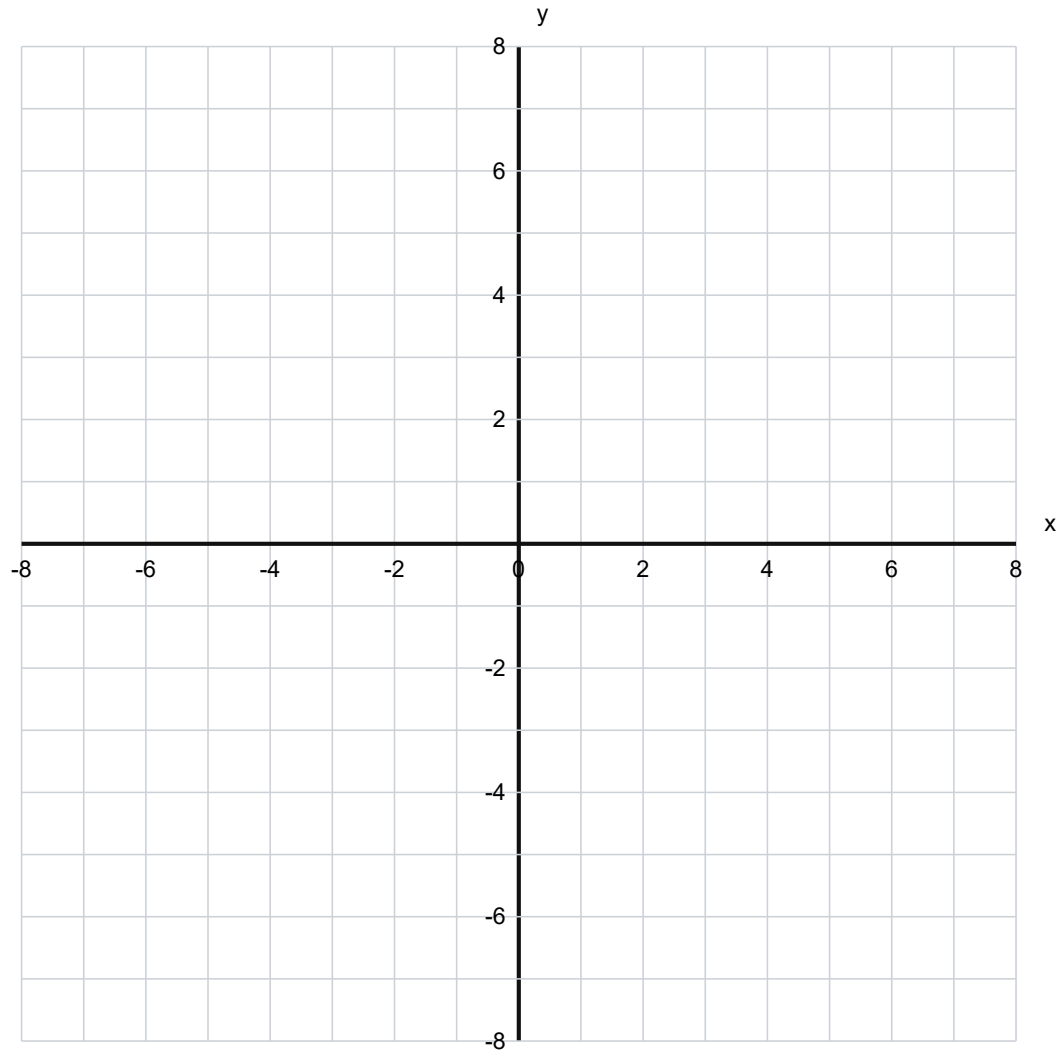
Given graph: f

Work space



Question 3: your graph

One unit per square.



Answers and work

Question 4

10 points

4 Problem 4

10 points

Let $R(x) = \frac{x^2 - x - 2}{x^2 + x - 6}$.

(a) Simplify $R(x)$ and state its domain. Show your work.

(b) Solve $R(x) = \frac{1}{2}$.

(c) Let $S(x) = \frac{x+1}{x+3}$. Do R and S have the same domain? Explain.

Work space

Answers and work

Question 5

6 points

5

Problem 5

6 points

Let $A(x) = x^2 - 2x$ and $B(x) = 2x - 1$.

(a) Find and simplify $A(B(x))$ and $B(A(x))$.

(b) Find all real x -values for which $A(B(x)) = B(A(x))$. Justify your answer.

Answers and work

Question 6

14 points

6

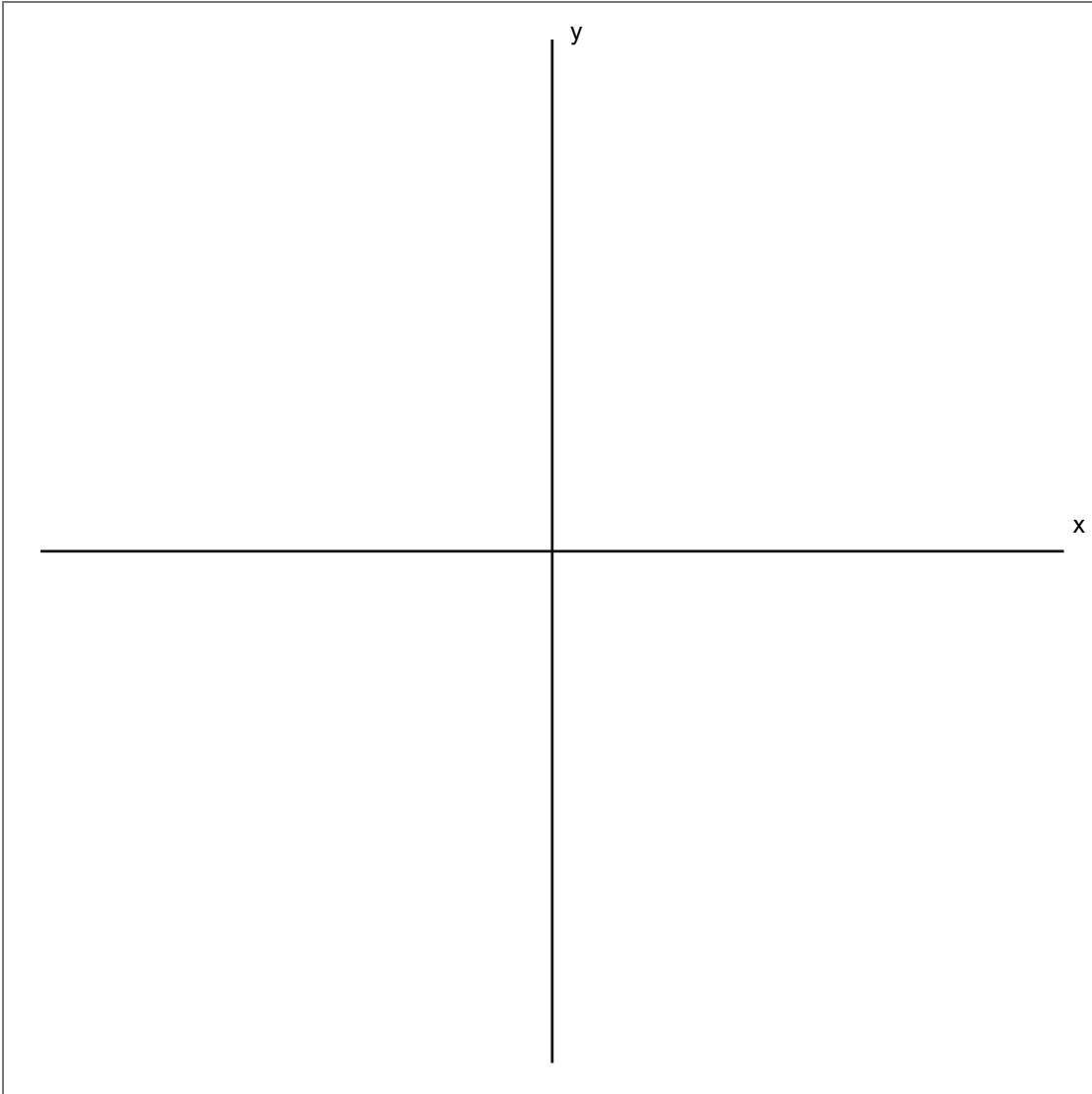
Problem 6

14 points

A degree-5 polynomial P has exactly two distinct real zeros: -2 with multiplicity 2, and 1 with multiplicity 3. It passes through $(0, -8)$.

- (a) Write an equation for P in factored form. Show your work.
- (b) Sketch P on the provided axes. Label its intercepts.
- (c) State where $P(x) > 0$ and $P(x) < 0$. Explain your answer.

Question 6: your qualitative sketch



Answers and work

Question 7

8 points

7 Problem 7

8 points

Let $f(t) = \sqrt{t}$, $g(x) = 9 - x^2$, and $H(x) = \frac{f(g(x))}{g(x)-5}$.

- (a) Write H as one expression in x.
- (b) Find the domain in interval notation. Show why each excluded value is excluded.
- (c) Is x=3 allowed? Is x=2 allowed? Explain using the original expression.

Answers and work

Question 8

8 points

8

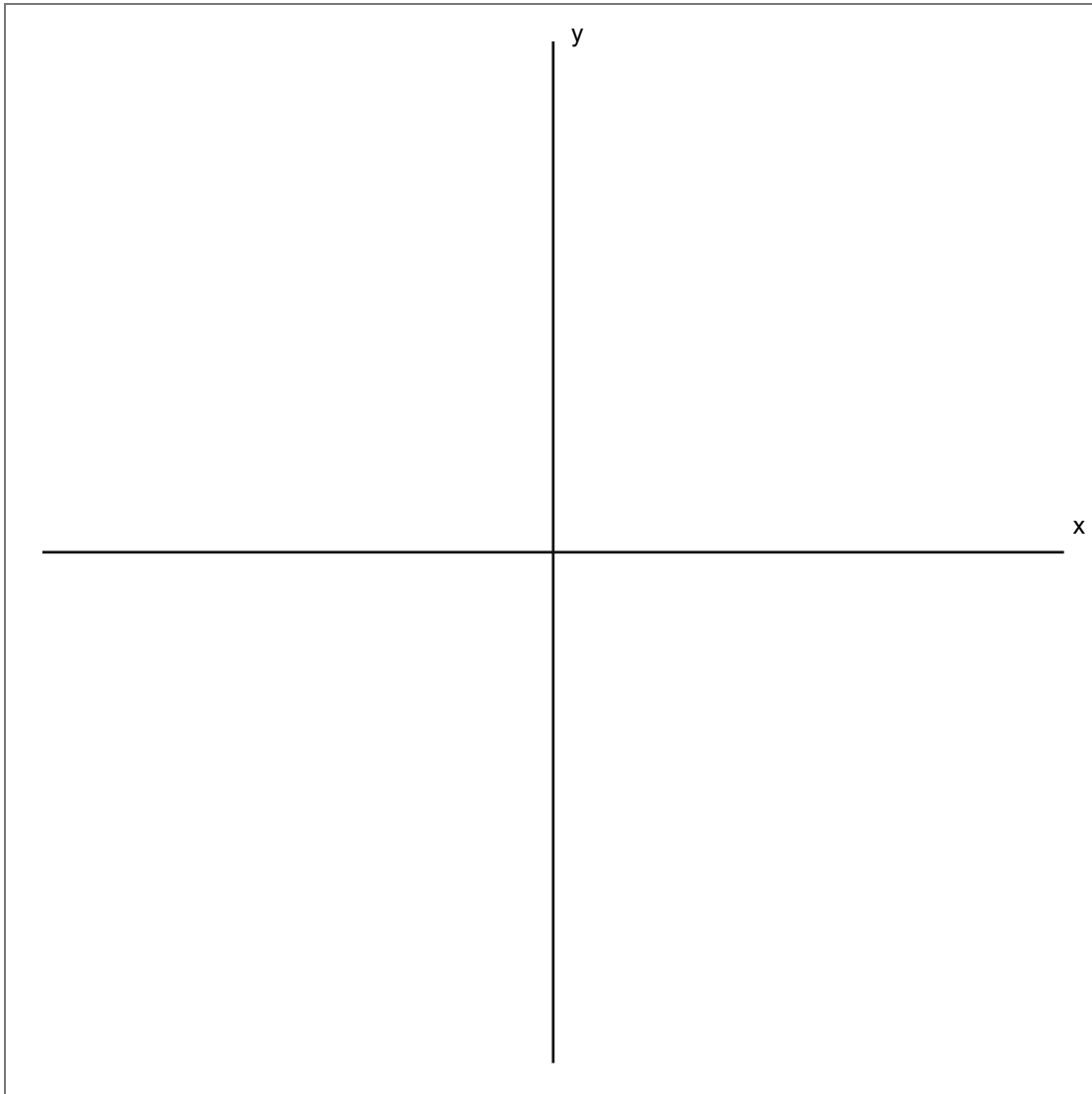
Problem 8

8 points

Let $R(x) = -(x + 2)(x - 1)^2(x - 3)$.

- (a) State the degree, leading coefficient, zeros, and multiplicities. Find the y-intercept.
- (b) Sketch R on the provided axes. Label its intercepts.
- (c) State the intervals where $R(x) > 0$ and $R(x) < 0$. Compare its end behavior with the polynomial in Question 6.

Question 8: your qualitative sketch



Answers and work

Question 9

10 points

9

Problem 9

10 points

A rectangular planting bed has perimeter 24 feet. Its width is w feet, with $2 \leq w \leq 6$, and its length is $12 - w$ feet. Its area is $A(w) = w(12 - w)$ square feet.

Soil delivery costs $C(a) = 8a$ dollars when $0 \leq a \leq 30$ square feet, and $C(a) = 240 + 5(a - 30)$ dollars when $a > 30$ square feet. There are no other charges.

- (a) Write the composition giving cost in terms of width. Explain the order using units.
- (b) Write that cost as a piecewise function of w , including the width intervals for each price rule. State the model domain.
- (c) Find the cost when $w=4$.
- (d) With a budget of \$260, find the greatest permitted width. Give an exact value and a decimal to the nearest hundredth. Show your work.

Answers and work

Question 10

8 points

10 Problem 10

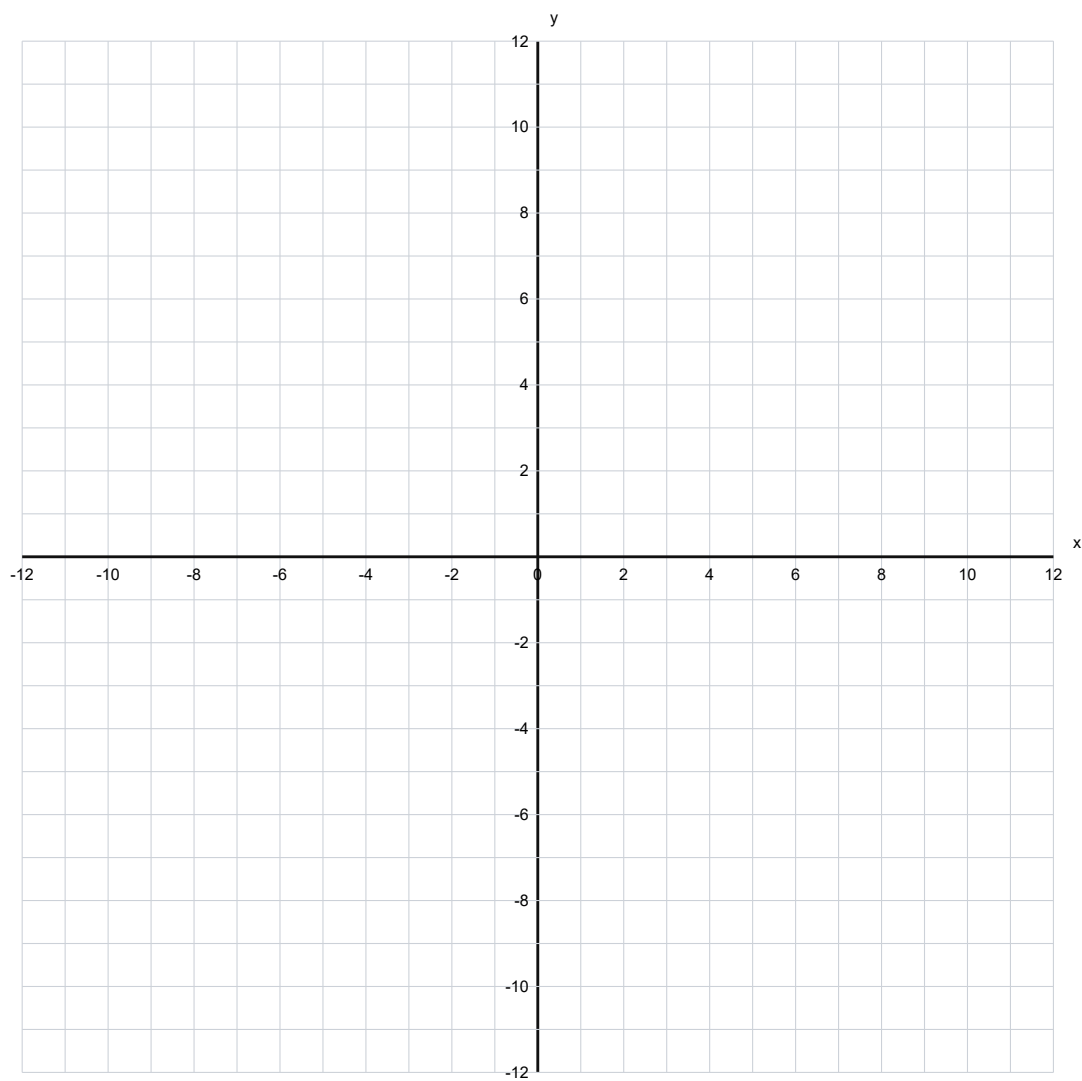
8 points

A square-root function has the form $r(x) = a\sqrt{x - h} + k$, domain $[1, \infty)$, range $(-\infty, 3]$, and passes through $(5, -1)$.

- (a) Find its equation. Show your work.
- (b) Graph r on the provided coordinate plane. Label its starting point and three other points.
- (c) State its x -intercept exactly and its interval of decrease.

Question 10: your graph

One unit per square.



Answers and work